北京化工大学

2022-2023-1 学期期末试卷 A

课程名称： 产品工程学 课程代码：**CHE41200T** 任课教授_____

班级:_____ 姓名：_____学号：_____

Note: please answer all questions ***in English***, and finish in **120 min**.

1. Please fill in the blanks with appropriate products (**30 points**).
   The chemical product in the *Chemical Product Engineering* refers to the functional products that are related to chemicals, chemical reactions or principles, and are used directly by the end consumers to satisfy their targeted purposes. The product may be a real or virtual one. Examples of virtual chemical products include (a)_____ and (b)_____. The real chemical products vary greatly from specialty chemicals that are chemicals with special functionality like (c)_____ and (d)_____, to formulated products that are composite with different components and functionality like (e)_____ and (f)_____, to devices that are related to specific chemical reactions/principles, like (g)_____ and (h)_____, to bio-based chemical products, like (i)_____ and (j)_____; and even to micro-structured products for which the micro-sized structure dominates their property and functionality like (k)_____and (l)_____. For the micro-structured products, their properties not only depend on (m)_____ that controls their thermodynamic property, but also on (n)_____ that controls their dynamic property. On the contrary, the bulk or commodity chemicals that are usually produced hundreds of thousands tones annually do not belong to the scope of the chemical product design like (o)_____ and (p)_____.
2. **Warm baby bottle** (**30 points**)**.** Parents of babies often wish to heat milk at around 40℃ while travelling or while in locations where there is no easily available sources of hot water except bottled water. We propose making a device that consists of a sleeve filled with supersaturated sodium acetate tri-hydrate ($CH_3CO_2Na.3H_2O$). This material crystalizes at 57℃, but spontaneous nucleation does not occur above -25℃. The product is therefore stable when molten until crystallization is initiated. This can be done by bending a crack metal disk held in the solution. Once nucleated, heat is generated and the milk warmed. Please
   (1) Find the true needs and the key specifications of the product;
   (2) List the missing information necessary for the product design?
   (3) What is the trigger of the product?
   (4) What is the demerits and limitation of this product?
   (5) What is the potential substitutes or competitive product of this product?
3. **Please develop a new chemical product for COVID-19 (40 points**)
   COVID-19 is a frustrating epidemic, which has lasted for three years worldwide and led over millions of death tolls. The global scientists have been working hard to find effective ways to cure the disease or prevent its spreading. Now, as a new graduate, you are encouraged to initiate a new company in accordance with the national calls for "**mass entrepreneurship and innovation**" under the help of interest free-loan initiative. As the head of the company, you are planning to build up a new product development team to develop a chemical product that includes but not limited to a new Chinese drug, a chemical medicine, healthcare wares or special devices based on your expertise/experience, education background, interests, or market needs.
   1) Please list the main work to be conducted as a leader of the product team.
   2) Please list the main procedure and the objectives of each step.
   3) Please determine the true needs of your new product, and its key specifications.
   4) Draw a diagram sketch of your product to illustrate its working principle.
   5) What kinds of missing information should be collected further for the design of the product?
   6) Draw a Gantt-chart for the development of the new product including the investment plan.